How much RL relative to pretraining? Create-vs-elicit papers, lab disclosures, frontier estimates and our runs

verdict: elicit / narrows
verdict: create / expands
verdict: mixed
no create-vs-elicit test
our runs: LLM-equivalent RL-only to as-run
FLOPs from tokens (low to high)
FLOPs from GPU-h x MFU 3-30 %
GPU-hour ratio
stated ratio (arrow = lower bound)
frontier H2 2025-26: central
frontier H2 2025-26: range

frontier labs H2 2025-26
central 0.3-2 (dark), range 0.1-5 (light)
weakest source grade D

Frontier labs: graded estimates (this review)

2024 (o1 era) — grade C
H1 2025 — grade C
H2 2025 — grade D
2026 — grade D

Lab disclosures: GPU-hours or stated ratios

Olmo 3.1 Think 32B (RL +21 days) — no test
DeepSeek-R1: RL 41K, all R1 work 147K H800-h vs V3 2,783K — no test
DeepSeek-V3.2 post-training — no test
Llama-3.1-Nemotron-Ultra-253B reasoning RL — no test
Reflection Beam (501B, 23B active) — no test
Composer 1.5 (base pretrained elsewhere) — no test

Yue et al.'s RL models

Qwen-2.5-7B-SimpleRL-Zoo — elicit
Qwen-2.5-14B-SimpleRL-Zoo — elicit
Qwen-2.5-32B-SimpleRL-Zoo — elicit
Llama-3.1-8B-SimpleRL-Zoo — elicit
Qwen2.5-Math-7B-Oat-Zero — elicit
DAPO-Qwen-32B — elicit
CodeR1-Zero-Qwen2.5-7B-12k-832 — elicit
DeepCoder-14B-Preview — elicit
Qwen2.5-VL-7B-Instruct + EasyR1 GRPO, Geometry3K — elicit
Qwen2.5-7B + GRPO, 150 steps (algorithm comparison; PPO-fami... — elicit
Qwen2.5-7B + GRPO, 450 steps (longest run) — elicit
Qwen2.5-7B + PPO, 150 steps (critic) — elicit
Qwen2.5-7B + DAPO, 150 steps (3-6x over-generation) — elicit
Qwen2.5-7B + GRPO n = 32, 220 steps — elicit
Qwen2.5-Math-7B-Instruct (SFT + GRPO with RM) — elicit
Magistral-Medium-2506 — *ratio not computable (size / pretraining / steps undisclosed)*

Other papers' own RL runs

100M decoder from scratch (GSM-Infinite), RL 1.05B-20B token... — mixed
MiniMax-M1 (full RL) — no test
ProRL v2 (3,000 steps) + BroRL (+191 steps, 512 rollouts per... — mixed
AceReason-Nemotron-7B (math-RL runs, GPU-h axis) — create
Open-Reasoner-Zero-32B — no test
DeepSeek-R1-Zero — no test
JustRL-DeepSeek-1.5B (4,380 steps) — no test
Nemotron-Research-Reasoning-Qwen-1.5B v1 (ProRL, 2,000 steps) — mixed
INTELLECT-3 (GLM-4.5-Air base) — no test
Skywork-OR1-32B (released at step 1,000) — no test
Olmo 3 Think 32B (final RL) — no test
ScaleRL on Llama-4-Scout (17B active), 50k GB200-h — no test
Olmo 3 7B RL-Zero (math) — create
Polaris-4B-Preview — no test
Qwen2.5-Math-1.5B + GRPO on one example — elicit
Qwen2.5-Math-7B + NSR / PSR / GRPO (MATH) — mixed
Phi-4-reasoning-plus (RL stage, 90 steps) — no test
Qwen2.5-Math-7B + GRPO-N (~42 steps) — mixed
Llama-3.1-8B-Instruct + RFT + DAPO on compositions — create
Qwen2.5-Math-7B + GRPO, 300 steps (random / true reward) — elicit
Ling-2.5-1T zero-RL (3,600 steps) — *ratio not computable (size / pretraining / steps undisclosed)*
Qwen3 reasoning RL (235B-A22B: 3,995 queries, 170 steps) — *ratio not computable (size / pretraining / steps undisclosed)*

Our runs (9.56 M lean_staten, from scratch)

trajectory: 3 seed(s), 8-8 EI rounds
trajectory-cap6: 3 seed(s), 8-8 EI rounds
rl-continue: 3 seed(s), 16-16 EI rounds
rl-continue-cap6: 3 seed(s), 16-16 EI rounds
rl-from-ckpt start p12000: 3 seed(s), 8-8 EI rounds
rl-from-ckpt start p1600: 3 seed(s), 8-8 EI rounds
rl-from-ckpt start p16000: 3 seed(s), 8-8 EI rounds
rl-from-ckpt start p5000: 3 seed(s), 8-8 EI rounds
create-or-elicit dose1: 3 seed(s), 16-16 EI rounds
create-or-elicit full: 3 seed(s), 16-16 EI rounds
create-or-elicit ko: 6 seed(s), 16-24 EI rounds

10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2}

RL compute / pretraining compute of the same model (log scale; 1 = equal)

Sources, assumptions and every value: rl_compute_share/compute_table.csv (the table view), ledger/*.tsv, frontier_ranges.csv, our_runs.csv.
Pretraining = 6ND; RL = 2N per generated token + 6N per trained token (+2N per extra forward pass); N cancels when RL runs on the pretrained model itself.